

Safety Data Sheet(SDS)

1. Identification of the substance/mixture and of the company/undertaking

1) Product identifier : AA-B

2) Relevant identified uses of the substance or mixture and uses advised against

○ Relevant identified uses

48.Others (undefined)

○ Uses advised against

Recommended use only

3) Supplier information

○ Company name [Manufacture]

Company : LG Chem, Ltd.

Address : 501, Sandanjungang-ro, Yeosu-si, Jeollanam-do, Republic of Korea

Emergency number : 0618055130/5511/5611

2. HAZARD IDENTIFICATION

1) Hazard classification

- Flammable liquids Category 3
- Acute toxicity(Oral) Category 3
- Acute toxicity(Dermal) Category 3
- Acute toxicity(Inhalation:Vapours) Category 4
- Skin corrosion/irritation Category 1A
- Specific target organ toxicity single exposure Category 3(Respiratory tract irritation)
- Hazardous to the aquatic environment, short-term (acute) Acute 1

2) Allocation label elements

Hazard pictograms



Signal word

- DANGER

Hazard statements

H226 Flammable liquid and vapour
H301 Toxic if swallowed
H311 Toxic in contact with skin
H314 Causes severe skin burns and eye damage
H332 Harmful if inhaled
H335 May cause respiratory irritation
H400 Very toxic to aquatic life

Precautionary statements

- Prevention

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233 Keep container tightly closed.
P240 Ground and bond container and receiving equipment.
P241 Use facilities such as explosion-proof electricity, ventilation and lighting.
P242 Use nonsparking tools.
P243 Take action to prevent static discharges.
P260 Do not breathe mist/vapours/spray.
P261 Avoid breathing mist/vapours/spray.
P264 Wash face, skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a wellventilated area.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

- Response

P301+P310 If you swallow: immediately receives medical institutions and doctors' consultation.
P301+P330+P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomition.
P302+P352 If you get on your skin: Wash with a large amount of water.
P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately receive medical institutions and doctors' consultation.
P312 If you feel uncomfortable, receive medical institutions and doctors' consultation.
P321 If you eat or inhale the substance, do not take artificial breath in the oral cavity, and use appropriate breathing medical equipment.,When contacting with substances, make a treatment such as rinsing the skin and eyes in the water flowing more than 20 minutes.
P330 Rinse mouth.
P361+P364 Take off immediately all contaminated clothing and wash it before reuse.
P363 Wash contaminated clothing before reuse.
P370+P378 In case of fire: Use carbon dioxide,Water,Alcoholic to extinguish.
P391 Collect spillage.

- Storage

P403+P233 Store in a wellventilated place. Keep container tightly closed.

P403+P235 Store in a wellventilated place. Keep cool.

P405 Store locked up.

- Disposal

P501 Dispose of contents and containers according to the legislation of the waste

3) Other hazards

- No data available

○ Product NFPA Level

Health	Flammability	Reactivity
3	2	2

(※ 0 = Stable , 1 = Low , 2 = Medium , 3 = High , 4 = Very High)

3. Composition/Information on ingredients

Components	Common name	CAS No.	PCT(wt%)
Acrylic acid	Acrylic acid	79-10-7	100

4. FIRST AID MEASURES

1) Following eye contact

- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Seek immediate medical assistance.

2) Following skin contact

- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat.
- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat. Do not remove clothing adhering to the skin in case of severe burns, seek medical aid.
- For minor skin contact, avoid spreading material on unaffected skin.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Remove and isolate contaminated clothing and shoes.
- Seek immediate medical assistance.
- Wash skin with soap and water.

3) Following inhalation

- Administer oxygen if breathing is difficult.

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Effects of contact or inhalation may be delayed.
- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Give artificial respiration if victim is not breathing.
- Keep victim warm and quiet.
- Move to fresh air.
- Seek immediate medical assistance.

4) Following ingestion

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medical assistance.

5) Advice to physician

- Effects of contact or inhalation may be delayed.
- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

5. FIRE FIGHTING MEASURES

1) Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media
 - CO₂.
 - Dry chemical.
 - For mixtures containing alcohol or polar solvent: Alcohol-resistant foam.
 - Regular foam.
 - Use alcohol foam, carbon dioxide, or water spray when fighting fires involving this material.
 - Use dry sand or earth to smother fire.
 - Water spray.
- Unsuitable extinguishing media
 - Direct water.
 - High-pressure water.

2) Special hazards arising from the substance or mixture

- Pyrolytic product
 - During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.
- Risk of fire and explosion
 - Can form explosive mixtures at temperatures at or above the flashpoint.
 - Containers may explode when heated.
 - Flammable/combustible material.

- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- May ignited from heat, friction or contamination.
- May violently polymerize and result in fire and explosion.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.

○ Other

- Contact may cause severe burns to skin and eyes.
- Fire may produce irritating, corrosive and/or toxic gases.
- May cause toxic effects if inhaled or ingested/swallowed.
- Vapors may cause dizziness or asphyxiation without warning.

3) Special protective equipment for firefighters

- Cautions ; Most of liquids are lighter than water.
- Dike fire-control water for later disposal; do not scatter the material.
- Evacuate area and fight fire from a safe distance.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.
- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks: Do not get water inside containers.
- Fire involving Tanks: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.
- Fire involving Tanks: For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks).
- Move containers from fire area if you can do it without risk.
- Rescuers should put on appropriate protective gear.

6. ACCIDENTAL RELEASE MEASURES

1) Health considerations and protective equipment

- A vapor suppressing foam may be used to reduce vapors.
- All equipment used when handling the product must be grounded.
- Clean up spills immediately, observing precautions in Protective Equipment section.
- Cover with plastic sheet to prevent spreading.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.

- Do not touch or go near exposed material.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- Fully encapsulating, vapor protective clothing should be worn for spills and leaks with no fire.
- Please note that materials and conditions to be avoided.
- Stop leak if you can do it without risk.
- The very fine particles can cause a fire or explosion, eliminate all ignition sources.

2) Environmental precautions

- Keep out of waterways.
- Prevent entry into waterways, sewers, basements or confined areas.
- Runoff may cause pollution.

3) For cleaning up

- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.
- Absorb spill with inert material (e.g., dry sand or earth), then place in a chemical waste container.
- Absorb the liquid and scrub the area with detergent and water.
- Dike and collect water used to fight fire.
- Large Spill: Dike far ahead of liquid spill for later disposal.
- Small Spill: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.
- Use clean non-sparking tools to collect absorbed material.

7. HANDLING AND STORAGE

1) Precautions for safe handling

- All equipment used when handling the product must be grounded.
- Avoid breathing vapors from heated material.
- Avoid prolonged or repeated contact with skin.
- CAUTION: High temperature.
- Caution: Heat.
- DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION;.
- Do not enter storage area unless adequately ventilated.
- Follow all MSDS/label precautions even after container is emptied because they may retain product residues.
- Handling refer to engineering control/personal protection section.
- Loosen closure cautiously before opening.
- Measure atmospheric oxygen concentration and ventilate the area during the operation since low-closed area can cause oxygen deficiency.
- Please note that materials and conditions to be avoided.
- Use care in handling/storage.
- Use only in a well-ventilated area.

2) Conditions for safe storage (including any incompatibilities)

- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed of.
- Keep away from food and drinking water.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

1) Chemical exposure limits, Biological exposure standard

Components	Occupational exposure limits	ACGIH	Biological standard
Acrylic acid	TWA : 2.0ppm STEL : Not applicable	TWA : Not applicable STEL : Not applicable	Not applicable

2) Appropriate engineering controls

- Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.
- If user operations generate dust, fume, or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.
- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.

3) Personal protection equipment

- Respiratory protection
 - In case of insufficient oxygen (<19.6%), wear a supplied air mask or self-contained respirator.
 - Wear breathing protection, which needs a confirmation from the Korea Occupational Safety and Health Agency.
- Eye protection
 - Wear suitable protective goggles and face shields.
- Hand protection
 - Wear suitable protective gloves.
- Body protection
 - Wear suitable protective clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Colorless Liquid
Physical state	Liquid
Colour	Below 16 °C, solid; Above 16 °C, colorless liquid
Odour	Odor
Odour threshold	0.094 (uL / L)
pH	2.5
Melting point/freezing point	14 °C
Initial boiling point and boiling range	141°C
Flash point	50°C
Evaporation rate	Not available
Flammability(solid, gas)	Flammable liquid

Upper/lower flammability or explosive limits	8 / 2.4%
Vapour pressure	413 Pa (20 ° C)
Solubility(ies)	100 g / 100 ml (25 ° C, availability)
Vapour density	2.5
Relative density	1.05
n-octanol/water partition coefficient	log Kow= 0.35 at 25 °C (OECD TG 107)
Auto ignition temperature	395 °C
Decomposition temperature	Not available
Viscosity	No data available
Molecular weight(mass)	72.06

10. STABILITY AND REACTIVITY

1) Stability and hazardous reactivity

- Can form explosive mixtures at temperatures at or above the flashpoint.
- Contact may cause severe burns to skin and eyes.
- Containers may explode when heated.
- Fire may produce irritating and/or toxic gases.
- Fire may produce irritating, corrosive and/or toxic gases.
- Flammable/combustible material.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- May cause toxic effects if inhaled or ingested/swallowed.
- May ignited from heat, friction or contamination.
- May violently polymerize and result in fire and explosion.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may cause dizziness or asphyxiation without warning.
- Vapors may form explosive mixtures with air.
- Vapors may travel to source of ignition and flash back.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.

2) Conditions to avoid

- Contact may cause severe burns to skin and eyes.
- Containers may explode when heated.
- Fire may produce irritating, corrosive and/or toxic gases.
- Flammable/combustible material.
- Heat, contamination.
- Heat.
- Ignition source(heat, spark, flame, etc.).
- May cause toxic effects if inhaled or ingested/swallowed.

- May ignited from heat, friction or contamination.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may cause dizziness or asphyxiation without warning.
- Vapors may travel to source of ignition and flash back.

3) Incompatible materials

- Combustibles, reducing material.
- Contact may cause severe burns to skin and eyes.
- Containers may explode when heated.
- Fire may produce irritating, corrosive and/or toxic gases.
- Flammable/combustible material.
- May cause toxic effects if inhaled or ingested/swallowed.
- May ignited from heat, friction or contamination.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Vapors may cause dizziness or asphyxiation without warning.
- Vapors may travel to source of ignition and flash back.

4) Hazardous decomposition products

- Corrosive/toxic fume.
- During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.
- Heat.
- Ignition source(heat, spark, flame, etc.).
- Irritating and/or toxic gas.
- Irritating, corrosive and/or toxic gas.

11. TOXICOLOGICAL INFORMATION

1) Exposure route information

- Inhalation
 - Harmful if inhaled
 - May cause respiratory irritation
- Skin Contact
 - Causes severe skin burns and eye damage
 - Toxic in contact with skin
- Eye Contact
 - Not applicable
- Ingestion
 - Toxic if swallowed

2) Health hazard information

○ Acute toxicity

● Acute toxicity(Oral)

Korea Ministry of Environment(Category 3 : 100mg/kg) - Acute toxic substances for humans(2020-1-983), Accident contrast substance(17)

● Acute toxicity(Dermal)

Korea Ministry of Environment(Category 3 : 300mg/kg) - Acute toxic substances for humans(2020-1-983), Accident contrast substance(17)

● Acute toxicity(Inhalation:Gases)

Korea Ministry of Environment(Category 4 : 4500ppm) - Acute toxic substances for humans(2020-1-983), Accident contrast substance(17)

● Acute toxicity(Inhalation:Vapours)

LC50 10.75 ~ 19.49 mg/ℓ 4 hr Test species: Rat ((vapor))

● Acute toxicity(Inhalation:Dust/mist)

No data available

○ Skin corrosion/irritation

Korea Ministry of Environment(Category 1A) - Acute toxic substances for humans(2020-1-983), Accident contrast substance(17)

○ Serious eye damage/eye irritation

No data available

○ Respiratory sensitization

No data available

○ Skin sensitization

No irritation observed in skin sensitization test on guinea pigs, Source: ECHA

○ Carcinogenicity

3 (IARC)

A4 (ACGHI)

○ Germ cell mutagenicity

In vitro gene mutation test using mammalian cultured cells, negative regardless of the presence of metabolic activation system OECD TG 476, GLP In vitro unscheduled DNA synthesis test using mammalian hepatocytes, negative in the absence of metabolic activation system OECD TG 482, GLP In vivo chromosome aberration test using mammalian bone marrow cells, negative OECD TG 475, GLP In vivo dominant lethality test using mice, negative GLP, Source: ECHA

○ Reproductive toxicity

As a result of the 2nd generation oral reproductive toxicity test using rats, NOAEL(P)=240mg/kg bw/day, NOAEL(F1, F2)=53mg/kg bw/day, no clinical symptoms related to toxicity were observed, hydrocephalus was observed at a concentration of 500ppm in the F2 generation (OECD TG 416, GLP). As a result of the 1st generation oral reproductive toxicity test using rats, NOAEL(P)=83mg/kg bw/day, NOAEL(F1)=250mg/kg bw/day, no clinical symptoms related to toxicity and death were observed, liver and kidney weights increased at intermediate and low concentrations in females of the P generation,

but there were no changes in body weight and organ weights, food and water consumption at high concentrations, and there were no changes in body weight and organ weights in the F1 generation (OECD TG 415). As a result of the inhalation developmental toxicity test using rabbits, no embryotoxicity or teratogenic effects were observed, and no deaths were observed, at 25ppm. No clinical symptoms observed, nasal congestion symptoms observed at 75 ppm, NOAEL (teratogenicity) $\geq$ 0.673 mg/L air, NOAEL (maternal/developmental toxicity) = 0.075 mg/L air (OECD TG 414, GLP) In the inhalation developmental toxicity test using rats, no teratogenic effects were observed, no deaths were observed, toxic effects were observed at 120 ppm (decreased body weight and uterine weight after autopsy, and decreased food consumption), and significant toxic effects were observed in pregnant individuals at 360 ppm (decreased body weight and food consumption, and irritation symptoms were observed), NOAEL (teratogenicity) $\geq$ 1.08 mg/L air, NOAEL (maternal/developmental toxicity) = 0.12 mg/L air (OECD TG 414, GLP), Source: ECHA

- Specific target organ toxicity single exposure

Korea Ministry of Environment(Category 3(Respiratory tract irritation)) - Acute toxic substances for humans(2020-1-983), Accident contrast substance(17)

- Specific target organ toxicity repeated exposure

Oral: 83 mg/kg bw/day/90 days (NOAEL, rat), Source: Chemical Substance Information Processing System (Information Summary)

- Aspiration hazard

No data available

12. ECOLOGICAL INFORMATION

1) Aquatic toxicity

- Fish

Korea Ministry of Environment(Hazardous to the aquatic environment, short-term (acute) Acute 1) - Ecotoxic substances(2020-1-983), Accident contrast substance(17)

- Crustacea

Korea Ministry of Environment(Hazardous to the aquatic environment, short-term (acute) Acute 1) - Ecotoxic substances(2020-1-983), Accident contrast substance(17)

- Aquatic algae

Korea Ministry of Environment(Hazardous to the aquatic environment, short-term (acute) Acute 1) - Ecotoxic substances(2020-1-983), Accident contrast substance(17)

2) Persistence and degradation

- n-octanol water partition coefficient

0.36 log Kow, Source: ICSC

- Degradation

No data available

- Biodegradation

90 ~ 100% 9 days (biodegradability, EU Method C.4-A, GLP), Source: ECHA

3) Bioaccumulative potential

3.162 (QSAR), Source: ECHA

4) Mobility in soil

6 ~ 137 Koc (EPA OTS 796.2750), Source: ECHA

5) Other adverse effects

CrustaceansDaphnia magna: NOEC21d=12mg/L OECD TG 211, GLP, Source: ECHA

13. DISPOSAL CONSIDERATIONS

1) Disposal methods

- Recyclables must be collected in portable bins and disposed of legally according to national regulations.

2) Precautions (including disposal of contaminated container of package)

- Avoid direct spillage to rivers, lakes, soil and drains.

14. TRANSPORT INFORMATION

1) UN No. : 2218

2) Proper shipping name : ACRYLIC ACID, STABILIZED

3) Class or division : 8

4) Packing group : II

5) Marine pollutant : Yes

6) Special safety response for transportation or transportation measure :

Emergency measures in case of fire : F-E

Emergency measures in the effluent : S-C

- ADR

- Tunnel restriction code : D/E

- IMDG

- Marine pollutant : Yes

- Air transport(IATA)

- UN No. : 2218
- Proper shipping name : ACRYLIC ACID, STABILIZED
- Class or division : 8
- Packing group : II

15. REGULATORY INFORMATION

1) Occupational Safety and Health Act in Korea

Environment measure substance(more than 1%), Hazardous Substances(Substances Subject to Process Safety Reporting), Hazardous Substances Requiring Management(more than 1%), Substance exposure limits

2) Toxic Chemical Control Act in Korea

Accident contrast substance(more than 25%), Acute toxic substances for humans(more than 5%), Ecotoxic substances(more than 25%)

3) Safety Control of Dangerous Substances Act in Korea

Class 4) Second petroleum (water soluble)(Designated quantity:2000ℓ)

4) Wastes Control Act in Korea

Designated waste (Waste toxic substances)

-In case of disposal, it must be disposed of in accordance with Article 13 of the Waste Management Act.

5) Other regulations in KOREA and Abroad regulations

- ETC regulation
 - Not applicable
 - Not applicable
- Act on the registration and evaluation of chemicals
 - Existing Commercial Chemical Substances, Existing chemicals subject to registration

16. OTHER INFORMATION

1) Reference

2) Date of first issue : 2026-01-08

3) Revision date

- Revised date count : 1
- Last revised date : 2026-01-12
- Last revised history : Reflecting the latest regulations

4) Other